Prem Banker

New York, NY | premb@vt.edu | (571) 640-0878 | LinkedIn | Github | Portfolio

Experience

Software Engineer, New York Life Insurance (Fortune 100 Insurance) – New York — Aug 2025 – Present

Tech Stack: Golang, Java, Python, AWS, OPA/Rego, HuggingFace, Electron

- Built a sub-**5ms** ABAC authorization service using a sidecar architecture with **Golang, OPA and Rego**, validated under **500 req/s** load with centralized policy management via AWS S3 and Lambda.
- Architected Sidekick's AI backend from scratch – an internal tool for meeting transcription, speaker diarization, and actionable insight generation using locally hosted **HuggingFace** models on EC2 with full CI/CD pipelines.
- Led migration to an in-house business rules engine, delivering a full POC and migration plan that eliminates vendor lock-in, saving **hundreds of thousands** in annual licensing costs.
- Drove **AWS RDS** cost optimization by analyzing CPU usage metrics and downsizing non-prod instances with zero downtime, saving **$53,000+** annually.

Software Engineer, Obviant (Defense AI Startup, $7.1M Seed) – Arlington, VA — Dec 2024 – Jul 2025

Tech Stack: React, Next.js, Node.js, TypeScript, AWS, Python, Terraform, Pinecone, Elasticsearch, Redis

- Joined as the **fourth engineer** at an early-stage defense-tech startup, contributing to a **$7.1M seed round** through rapid full-stack product delivery.
- Built **Recon**, a natural-language intelligence tool powered by RAG and hybrid search (Pinecone + Elasticsearch) with query expansion, enabling customers to query defense data conversationally.
- Designed a master-replica data pipeline syncing **∼2M** rows from **RDS** to **Elasticsearch** and **Pinecone** via event triggers, handling complex multi-entity updates at scale with queuing and batch processing.
- Implemented **RBAC** for file-sharing with granular, Google Drive-like permissions, addressing critical customer security requirements.

Full Stack Developer, Quicko (∼2M customers, Fintech) – Ahmedabad, India — May 2021 – Jun 2023

Tech Stack: Java SpringBoot, Node.js, Angular, Flutter, DynamoDB, Oracle SQL, AWS, GCP, Docker, Jenkins

- Led MEET development, boosting revenue by **80%** through automation of scheduling, payments, and order lifecycle using Flutter, Node.js, and AWS.
- Optimized scheduling with **AWS CloudFront** caching and EventBridge, reducing latency from **4s to 200ms**.
- Built a Calendly-like booking system with Google Calendar **APIs and OAuth**, saving over **$10k** annually.
- Developed a mobile PWA for tax filing plugged into major brokerages, onboarding **500,000+** users in 2022.
- Built Clique, an affiliate platform using Angular and Java SpringBoot, driving **800,000+** sign-ups through partner-led growth.

Application Development Intern, EVOSYS – Ahmedabad, India — Jan 2021 – Apr 2021

Tech Stack: JavaScript, PL/SQL, MySQL, Java, Oracle Cloud Infrastructure

- Built Oracle Integration Cloud (OICS) workflows and Oracle Fusion data pipelines to automate transfer and SFTP delivery.

Achievements

- Won the Ripple Award for best project using Ripple Chain at **MorganHacks** 2024 with CarbonKarma, a blockchain platform helping companies achieve carbon-neutral goals.
- Won **LetsHack 1.0** by building a Smart Parking System prototype using deep learning and CNNs.

Skills

Languages: Java, Golang, TypeScript, JavaScript, Python, Dart, SQL, HTML, CSS

Frameworks: React, Next.js, Node.js, SpringBoot, Angular, Flutter, HuggingFace

Cloud & Infrastructure: AWS (Lambda, S3, RDS, EC2, CloudFront, EventBridge), GCP, Docker, Terraform, OPA

Databases & Search: DynamoDB, PostgreSQL, Oracle SQL, Pinecone, Elasticsearch, Redis

Education

Virginia Tech, M.Eng. in Computer Science — Aug 2023 – May 2025

Pandit Deendayal Energy University, B.Tech. in Computer Engineering — Jul 2017 – May 2021